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ASSIGNMENT -1

Subject:Computer Networks

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Q2: Write the difference between Hub, Switch & Router in Tabular Format.

Explain why router is the most intelligent & Hub is least intelligent device.

Comparison Table

Feature	Hub	Switch	Router
OSI Layer	Physical Layer (Layer 1)	DataLink Layer(Layer 2)	Network Layer (Layer 3)
Data Unit	Bits/Electrical Signals	Frames (MACAddresses)	Packets(IP-Addresses)
Collision Domain	Single collision domain for all ports	Separate collision domain per port	Separate collision domain per port
Broadcast Domain	Single broadcast domain	Single broadcast domain (extended via VLANs)	Separates broadcast domains
Intelligence	None (dumb repeater)	Moderate (MAC table learning)	High(Path optimization & IP routing)

Why Router is Most Intelligent & Hub is Least Intelligent

- **Hub (Least Intelligent):** A hub is a simple hardware repeater. It operates entirely blindly at the physical layer, taking incoming electrical signals on one port and blindly broadcasting them out to every other port without reading headers, MAC addresses, or data contents. It cannot filter traffic, leading to massive collisions and security risks.
- **Router (Most Intelligent):** A router operates at the network layer, making complex logical decisions. It inspects source and destination IP headers, maintains dynamic

routing tables, computes optimal paths using routing algorithms (like Ospf or BGP), manages network congestion, and bridges disparate network architectures (e.g., LAN to WAN).

Q3: What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also Explain difference between CSMA/CD & CSMA/CA.

Definition & Need

- **What are they?** Contention-based protocols where no station is assigned a fixed time slot or frequency channel. Any station can transmit data whenever it has a frame ready, provided the channel is free.
- **Why do we need them?** Fixed channel allocation protocols (like TDM or FDM) waste bandwidth when assigned nodes are silent. Random access protocols provide dynamic, on-demand channel sharing, maximizing throughput and flexibility in multi-access environments.

Categories of Random Access Protocols

1. ALOHA Protocols

- **Pure ALOHA:** Stations transmit frames whenever available. If a collision occurs (detected via lack of ACK), nodes wait for a random back-off time before retransmitting.
 - Example: Early satellite or radio packet networks.
- **Slotted ALOHA:** Time is partitioned into discrete slots. Nodes can only transmit at the exact beginning of a time slot, cutting the vulnerable collision window in half and doubling maximum throughput.

2. Carrier Sense Multiple Access (CSMA)

- Stations "listen before they talk" (Carrier Sensing). If the channel is idle, transmission begins; if busy, the station defers.
- Persistence Variations: 1-persistent (transmits immediately when idle), Non-persistent (waits a random time if busy), P-persistent (transmits with probability p when idle).

3. CSMA with Collision Detection (CSMA/CD)

- Used in wired networks. Nodes listen during transmission. If a collision is detected, transmission halts immediately, a jam signal is sent, and nodes wait a back-off period (Binary Exponential Backoff).
- Example: Traditional wired Ethernet (IEEE 802.3).

4. CSMA with Collision Avoidance (CSMA/CA)

- o Used in wireless networks where hidden nodes prevent reliable collision detection. It uses a handshake mechanism (RTS/CTS: Request to Send / Clear to Send) to reserve the channel before sending data.
- o Example: Wi-Fi (IEEE 802.11).

Difference: CSMA/CD vs. CSMA/CA

Feature	CSMA/CD (Collision Detection)	CSMA/CA (Collision Avoidance)
Primary Strategy	Reacts to collisions after they occur.	Prevents collisions before they happen.
Environment	Wired LANs (Ethernet).	Wireless LANs (Wi-Fi).
Mechanism	Jam signal +Binary Exponential Back-off.	RTS/CTS handshaking+ Interframespacing(IFS)+ Acknowledgments.
Efficiency	Higher efficiency in wired setups due to fast propagation delay.	Lower efficiency due to overhead of handshake packets.